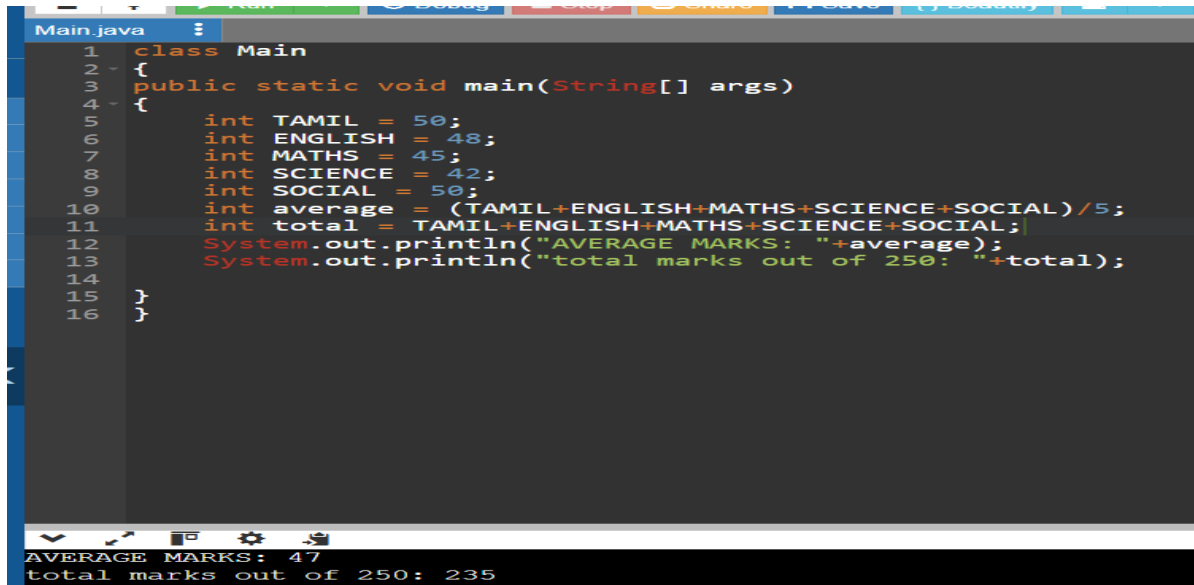


JAVA PROGRAM SUMS(11/7/26)

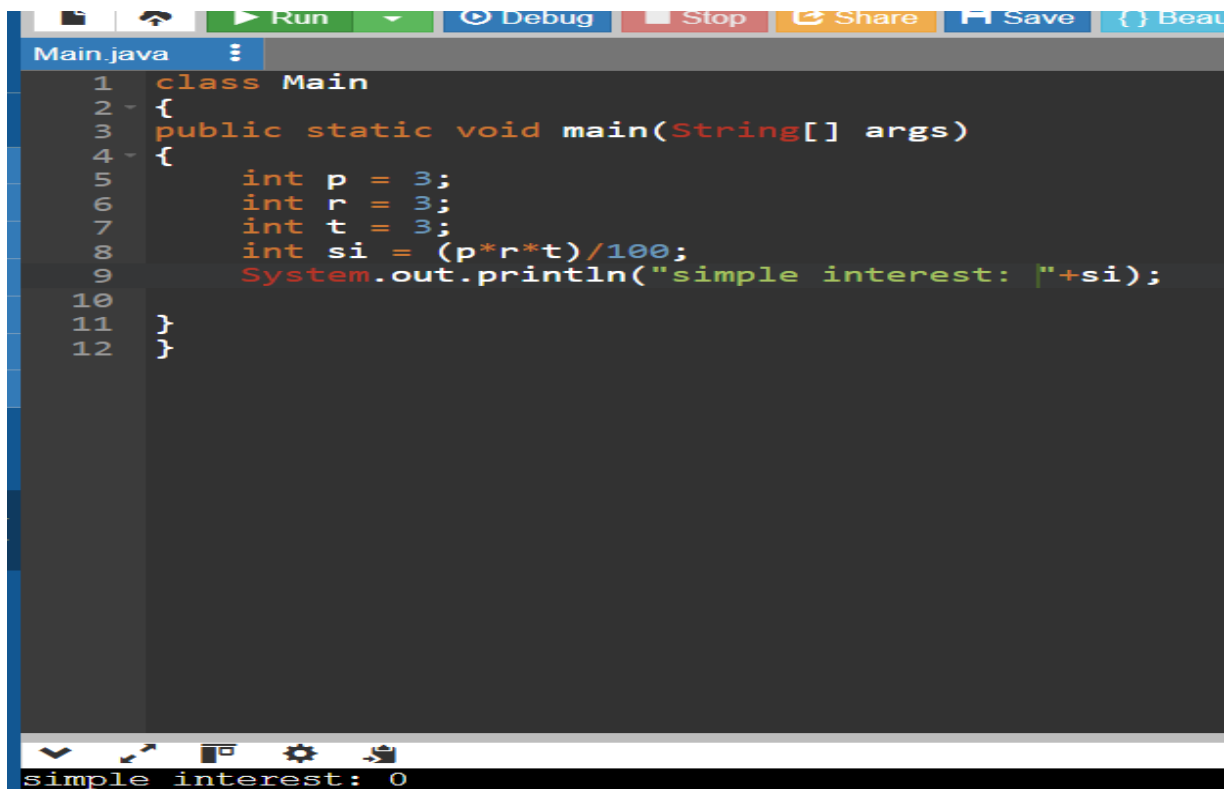
1. TO FIND AVERAGE:



```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int TAMIL = 50;
6         int ENGLISH = 48;
7         int MATHS = 45;
8         int SCIENCE = 42;
9         int SOCIAL = 50;
10        int average = (TAMIL+ENGLISH+MATHS+SCIENCE+SOCIAL)/5;
11        int total = TAMIL+ENGLISH+MATHS+SCIENCE+SOCIAL;
12        System.out.println("AVERAGE MARKS: "+average);
13        System.out.println("total marks out of 250: "+total);
14    }
15 }
16
```

AVERAGE MARKS: 47
total marks out of 250: 235

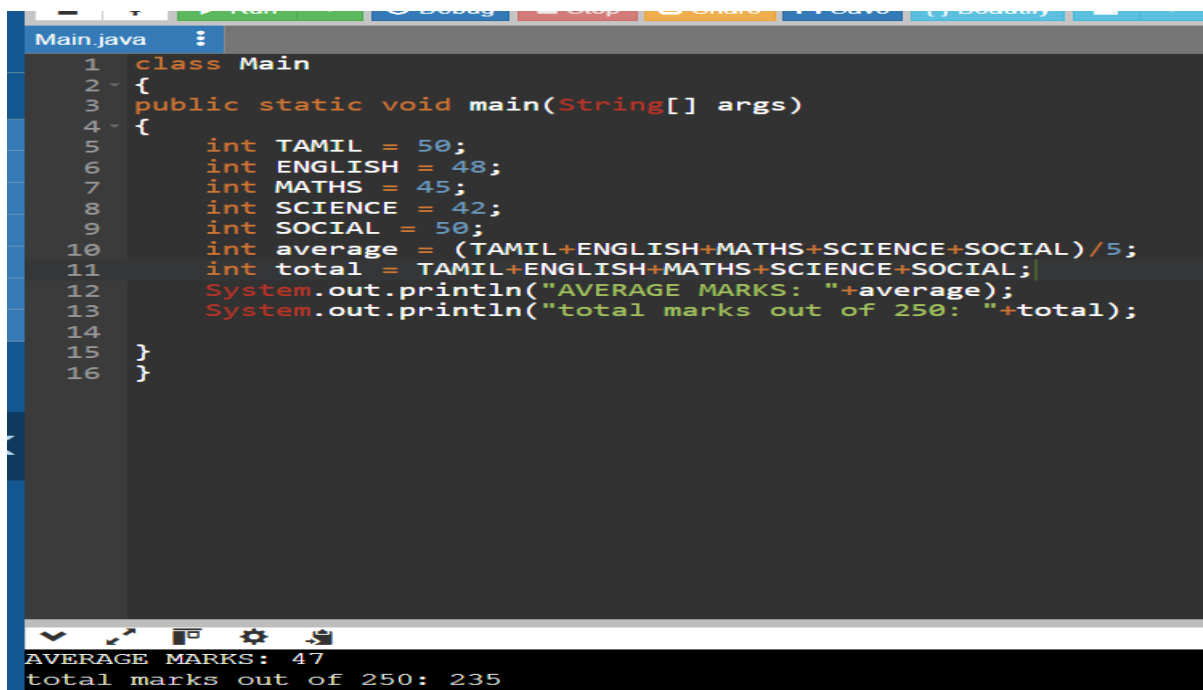
2. TO FIND SIMPLE INTEREST:



```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int p = 3;
6         int r = 3;
7         int t = 3;
8         int si = (p*r*t)/100;
9         System.out.println("simple interest: "+si);
10    }
11 }
12
```

simple interest: 0

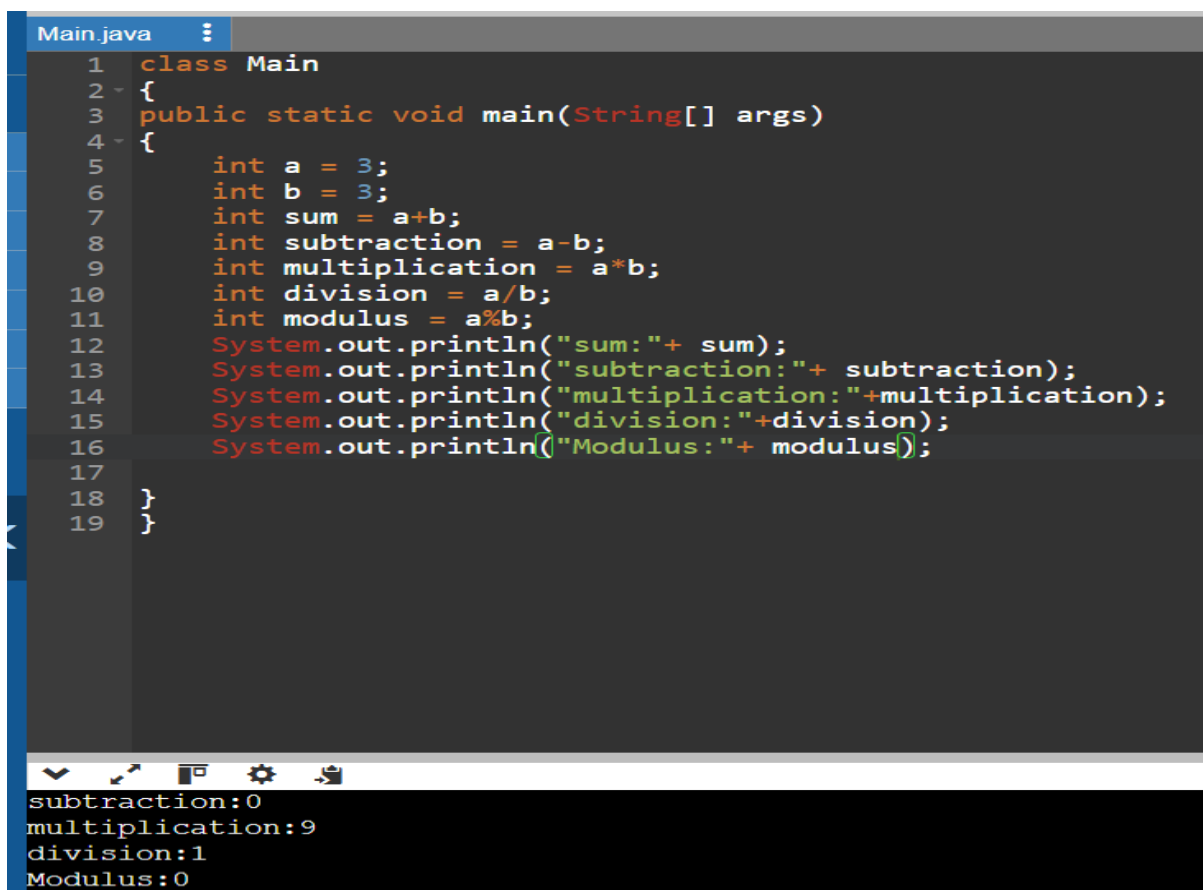
3. TO FIND STUDENT'S RESULT:



```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int TAMIL = 50;
6         int ENGLISH = 48;
7         int MATHS = 45;
8         int SCIENCE = 42;
9         int SOCIAL = 50;
10        int average = (TAMIL+ENGLISH+MATHS+SCIENCE+SOCIAL)/5;
11        int total = TAMIL+ENGLISH+MATHS+SCIENCE+SOCIAL;
12        System.out.println("AVERAGE MARKS: "+average);
13        System.out.println("total marks out of 250: "+total);
14    }
15 }
16 }
```

AVERAGE MARKS: 47
total marks out of 250: 235

4. ARITHMETIC OPERATORS:



```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int a = 3;
6         int b = 3;
7         int sum = a+b;
8         int subtraction = a-b;
9         int multiplication = a*b;
10        int division = a/b;
11        int modulus = a%b;
12        System.out.println("sum:"+ sum);
13        System.out.println("subtraction:"+ subtraction);
14        System.out.println("multiplication:"+multiplication);
15        System.out.println("division:"+division);
16        System.out.println("Modulus:"+ modulus);
17    }
18 }
19 }
```

subtraction:0
multiplication:9
division:1
Modulus:0

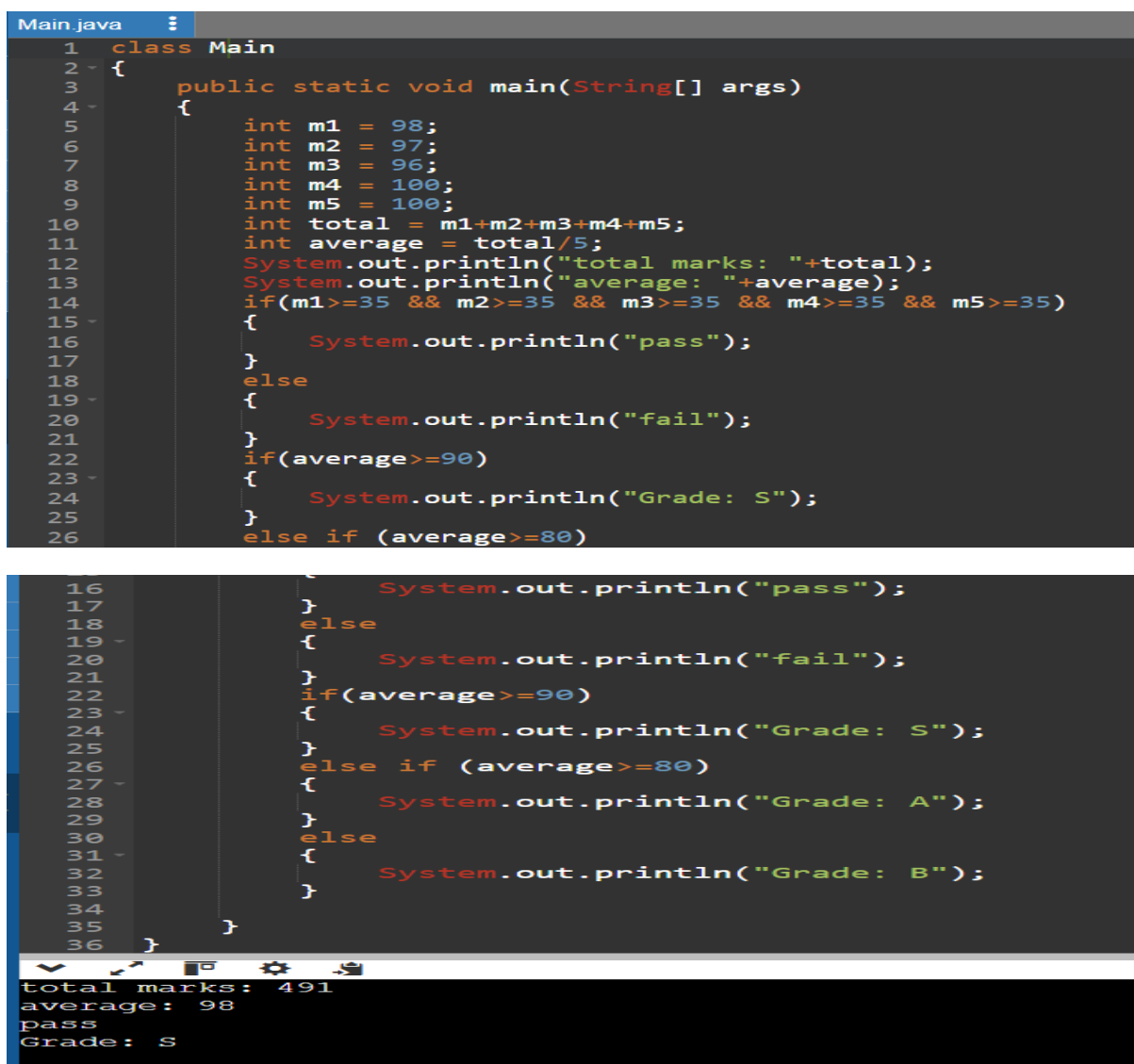
5. TO PRINT NAME:



```
class Main
{
    public static void main(String[] args)
    {
        System.out.println("Sivarangjenii");
    }
}
```

...Program finished with exit code 0
Press ENTER to exit console.

6.FIND STUDENTS RESULT USING IF , ELSE AND ELSE IF CONDITION:



```
1 class Main
2 {
3     public static void main(String[] args)
4     {
5         int m1 = 98;
6         int m2 = 97;
7         int m3 = 96;
8         int m4 = 100;
9         int m5 = 100;
10        int total = m1+m2+m3+m4+m5;
11        int average = total/5;
12        System.out.println("total marks: "+total);
13        System.out.println("average: "+average);
14        if(m1>=35 && m2>=35 && m3>=35 && m4>=35 && m5>=35)
15        {
16            System.out.println("pass");
17        }
18        else
19        {
20            System.out.println("fail");
21        }
22        if(average>=90)
23        {
24            System.out.println("Grade: S");
25        }
26        else if (average>=80)
27        {
28            System.out.println("Grade: A");
29        }
30        else
31        {
32            System.out.println("Grade: B");
33        }
34    }
35 }
36
```

total marks: 491
average: 98
pass
Grade: S

CLASS PRACTICE SUMS(14.07.2026) (PATTERNS USING LOOPS)

1.RIGHT TRIANGLE

PROGRAM EDITOR

```
1 public class Main
2 {
3     public static void main(String args[])
4     {
5         int n=4;
6         for(int i=1;i<=n;i++)
7         {
8             for(int j=1;j<=i;j++)
9             {
10                 System.out.print("* ");
11             }
12             System.out.println();
13         }
14     }
15 }
```

OUTPUT

```
*
* *
* * *
* * * *
```

2.LEFT TRIANGLE

```
1 public class Main
2 {
3     public static void main(String args[])
4     {
5         int n=4;
6         for(int i=1;i<=n;i++)
7         {
8             for(int j=1;j<=n-i;j++)
9             {
10                 System.out.print("* ");
11             }
12             System.out.println();
13         }
14     }
15 }
```

OUTPUT

```
* * *
* *
*
```

3.PYRAMID

```
1 public class Main
2 {
3     public static void main(String args[])
4     {
5         int n=4;
6         for(int i=1;i<=n;i++)
7         {
8             for(int j=1;j<=n-i;j++)
9             {
10                System.out.print(" ");
11            }
12            for(int k=1;k<=i;k++)
13            {
14                System.out.print("* ");
15            }
16            System.out.println();
17        }
18    }
19 }
```

OUTPUT

```
  *
 * *
* * *
* * * *
```

4.INVERTED PYRAMID

```
1 public class Main
2 {
3     public static void main(String args[])
4     {
5         int n=5;
6         for(int i=n;i>=1;i--)
7         {
8             for(int j=1;j<=n-i;j++)
9             {
10                System.out.print(" ");
11            }
12            for(int k=1;k<=(2*i-1);k++)
13            {
14                System.out.print("*");
15            }
16            System.out.println();
17        }
18    }
19 }
```

OUTPUT

```
*****
 *****
  *****
   ***
    *
```

5.DIAMOND

```
Main.java    Share

1 import java.util.Scanner;
2
3 public class Main
4 {
5     public static void main(String[] args)
6     {
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print("Enter number of rows: ");
10        int n = sc.nextInt();
11
12        // Upper half
13        for(int i = 1; i <= n; i++)
14        {
15            for(int j = 1; j <= n - i; j++)
16            {
17                System.out.print(" ");
18            }
19
20            for(int j = 1; j <= 2 * i - 1; j++)
21            {
22                System.out.print("*");
```

```
22         System.out.print("*");
23     }
24
25     System.out.println();
26 }
27
28 // Lower half
29 for(int i = n - 1; i >= 1; i--)
30 {
31     for(int j = 1; j <= n - i; j++)
32     {
33         System.out.print(" ");
34     }
35
36     for(int j = 1; j <= 2 * i - 1; j++)
37     {
38         System.out.print("*");
39     }
40
41     System.out.println();
42 }
```

```
42     }
43
44     sc.close();
45 }
46 }
```

```
Output
Enter number of rows: 5
  *
 ***
*****
*****
*****
*****
  *
 ***
  *
```

=== Code Execution Successful ===

6.HOLLOW SQUARE

```
Main.java
1 import java.util.Scanner;
2
3 class Main
4 {
5     public static void main(String[] args)
6     {
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print("Enter the size of the square: ");
10        int n = sc.nextInt();
11
12        for (int i = 1; i <= n; i++)
13        {
14            for (int j = 1; j <= n; j++)
15            {
16                if (i == 1 || i == n || j == 1 || j == n)
17                    System.out.print("* ");
18                else
19                    System.out.print("  ");
20            }
21            System.out.println();
22        }
23    }
24 }
```

```
Output
Enter the size of the square: 4
* * * *
*   *
*   *
* * * *
=== Code Execution Successful ===
```

7.HOLLOW RECTANGLE


```

Main.java
1 import java.util.Scanner;
2 public class Main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int rows = sc.nextInt();
8         int cols = sc.nextInt();
9
10        for(int i=1;i<=rows;i++)
11        {
12            for(int j=1;j<=cols;j++)
13            {
14                if(i==1 || i==rows || j==1 || j==cols)
15                    System.out.print("*");
16                else
17                    System.out.print(" ");
18            }
19            System.out.println();
20        }
21
22        sc.close();
23    }
24 }

```

```

4 6
*****
*   *
*   *
*****

```

8.SQUARE

```
Main.java
1 import java.util.Scanner;
2 public class Main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc=new Scanner(System.in);
7         int n=sc.nextInt();
8
9         for(int i=1;i<=n;i++)
10        {
11            for(int j=1;j<=n;j++)
12            {
13                System.out.print("*");
14            }
15            System.out.println();
16        }
17
18        sc.close();
19    }
20 }
```

4

9.RECTANGLE

```
1 import java.util.Scanner;
2 public class Main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc=new Scanner(System.in);
7         int rows=sc.nextInt();
8         int cols=sc.nextInt();
9
10        for(int i=1;i<=rows;i++)
11        {
12            for(int j=1;j<=cols;j++)
13            {
14                System.out.print("*");
15            }
16            System.out.println();
17        }
18
19        sc.close();
20    }
21 }
```

4 8

```
*****
*****
*****
*****
```

10.NUMBER PATTERN:

Write a program to print the below pattern?

1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

```
main.java
1 import java.util.Scanner;
2
3 public class Main
4 {
5     public static void main(String[] args)
6     {
7         Scanner sc = new Scanner(System.in);
8         int n = sc.nextInt();
9
10        for(int i = 0; i < n; i++)
11        {
12            int num = 1;
13
14            for(int j = 0; j <= i; j++)
15            {
16                System.out.print(num + " ");
17                num = num * (i - j) / (j + 1);
18            }
19
20            System.out.println();
21        }
22
23        sc.close();
24    }
25 }
```

5
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1

11. Write a program to print the following pattern

Sample Input:

Enter the number to be printed: 1

Max Number of time printed: 3

1

11

111

11

1

```

1 import java.util.Scanner;
2
3 public class Main
4 {
5     public static void main(String[] args)
6     {
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print("Enter the number to be printed: ");
10        int num = sc.nextInt();
11
12        System.out.print("Max Number of time printed: ");
13        int n = sc.nextInt();
14
15        for(int i = 1; i <= n; i++)
16        {
17            for(int j = 1; j <= i; j++)
18            {
19                System.out.print(num);
20            }
21            System.out.println();
22        }
23
24        for(int i = n - 1; i >= 1; i--)
25        {
26            for(int j = 1; j <= i; j++)
27            {
28                System.out.print(num);
29            }
30            System.out.println();
31        }
32
33        sc.close();
34    }
35 }

```

```

Enter the number to be printed: 1
Max Number of time printed: 3
1
11
111
11
1

```

12. Write a program to print the following pattern

Sample Input:

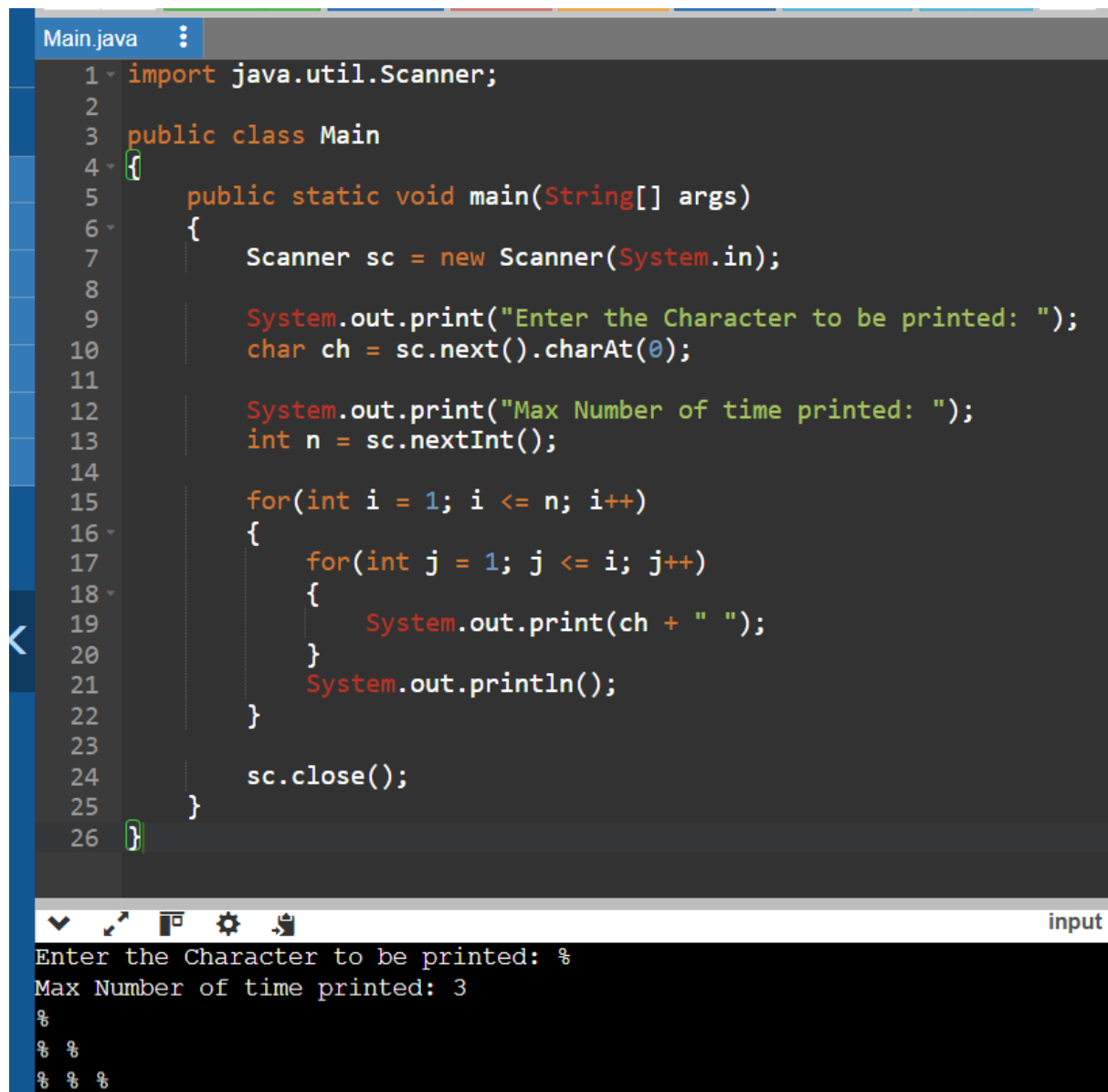
Enter the Character to be printed: %

Max Number of time printed: 3

%

% %

% % %



```
1 import java.util.Scanner;
2
3 public class Main
4 {
5     public static void main(String[] args)
6     {
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print("Enter the Character to be printed: ");
10        char ch = sc.next().charAt(0);
11
12        System.out.print("Max Number of time printed: ");
13        int n = sc.nextInt();
14
15        for(int i = 1; i <= n; i++)
16        {
17            for(int j = 1; j <= i; j++)
18            {
19                System.out.print(ch + " ");
20            }
21            System.out.println();
22        }
23
24        sc.close();
25    }
26 }
```

input

Enter the Character to be printed: %

Max Number of time printed: 3

%

% %

% % %

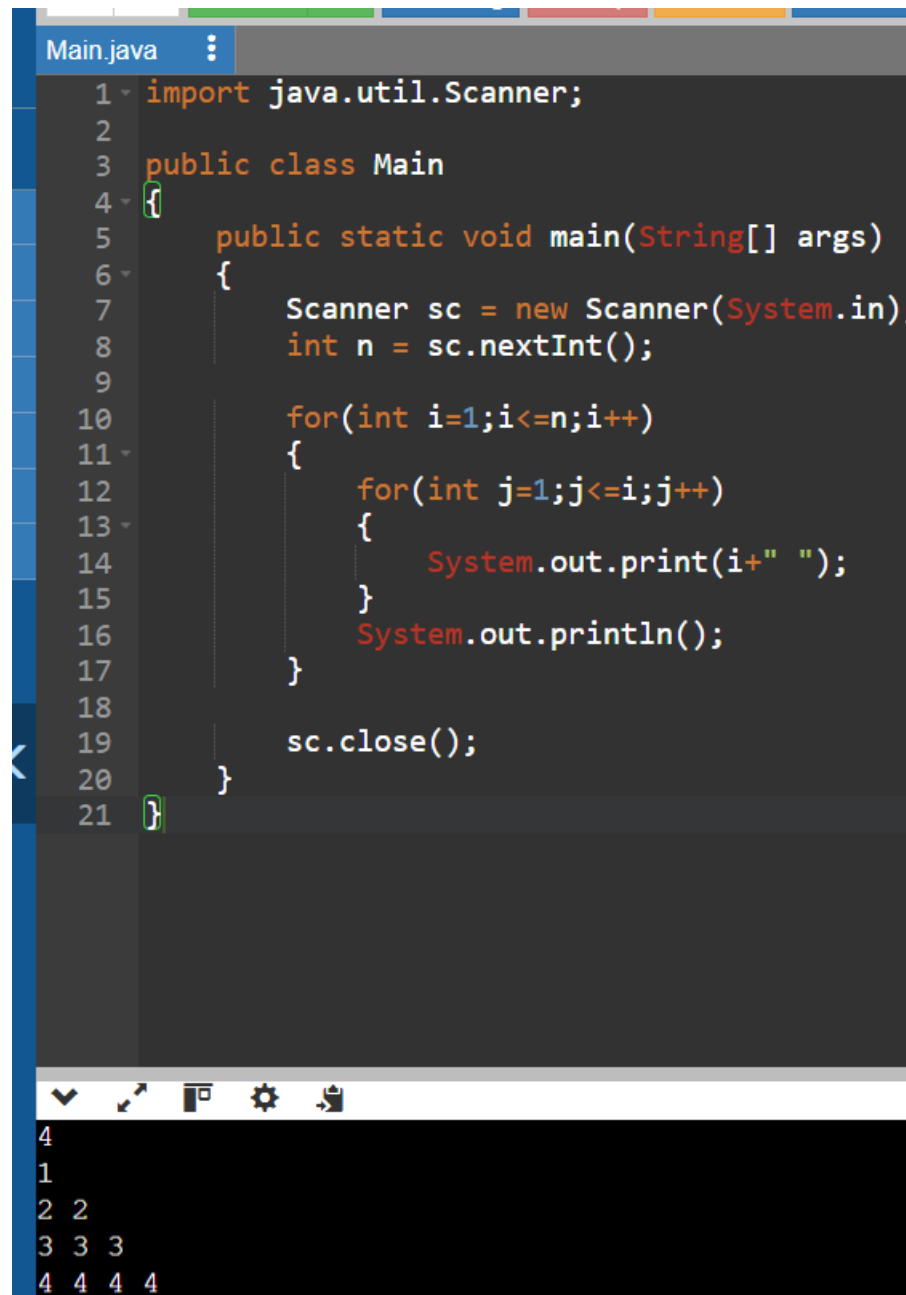
13. Write a program to print the below pattern

1

2 2

3 3 3

4 4 4 4



The screenshot shows a Java IDE with a file named 'Main.java'. The code is as follows:

```
1 import java.util.Scanner;
2
3 public class Main
4 {
5     public static void main(String[] args)
6     {
7         Scanner sc = new Scanner(System.in);
8         int n = sc.nextInt();
9
10        for(int i=1;i<=n;i++)
11        {
12            for(int j=1;j<=i;j++)
13            {
14                System.out.print(i+" ");
15            }
16            System.out.println();
17        }
18
19        sc.close();
20    }
21 }
```

Below the code editor, the output of the program is displayed in a terminal window, showing the pattern:

```
4
1
2 2
3 3 3
4 4 4 4
```


14. Write a program to print the below pattern

1

4 9

16 25 36

49 64 81 100

```
Main.java
1 import java.util.Scanner;
2
3 public class Main
4 {
5     public static void main(String[] args)
6     {
7         Scanner sc = new Scanner(System.in);
8         int n = sc.nextInt();
9
10        int num = 1;
11
12        for(int i=1;i<=n;i++)
13        {
14            for(int j=1;j<=i;j++)
15            {
16                System.out.print(num*num+" ");
17                num++;
18            }
19            System.out.println();
20        }
21
22        sc.close();
23    }
24 }
```

4
1
4 9
16 25 36
49 64 81 100

15. Write a program to print the below pattern

1

2 2

3 3 3

4 4 4 4

3 3 3

2 2

1

Main.java

```
1 import java.util.Scanner;
2
3 public class Main
4 {
5     public static void main(String[] args)
6     {
7         Scanner sc = new Scanner(System.in);
8         int n = sc.nextInt();
9
10        for(int i=1;i<=n;i++)
11        {
12            for(int j=1;j<=i;j++)
13            {
14                System.out.print(i+" ");
15            }
16            System.out.println();
17        }
18
19        for(int i=n-1;i>=1;i--)
20        {
21            for(int j=1;j<=i;j++)
22            {
23                System.out.print(i+" ");
24            }
25            System.out.println();
26        }
27
28        sc.close();
29    }
30 }
```

```
5
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
4 4 4 4
3 3 3
```

```
4 4 4 4
3 3 3
2 2
1
...Program finished with exit code 0
Press ENTER to exit console.
```

16.SUM OF FIBONACCI SERIES

```
Main.java
1 import java.util.Scanner;
2
3 public class Main {
4
5     static int fibonacciSum(int n) {
6         int a = 0, b = 1, c;
7         int sum = 0;
8
9         for (int i = 1; i <= n; i++) {
10             sum = sum + a;
11             c = a + b;
12             a = b;
13             b = c;
14         }
15
16         return sum;
17     }
18
19     public static void main(String[] args) {
20
21         Scanner sc = new Scanner(System.in);
22
23         int n = sc.nextInt();
24
25         System.out.println(fibonacciSum(n));
26
27         sc.close();
28     }
29 }
```

5
7

...Program finished with exit code 0
Press ENTER to exit console.

17.